URBANFIX AI / SYSTEM + ACCOUNTABILITY MAP

How citizen evidence becomes prioritized, assigned, and verifiably resolved

WORKING MVP | HUMAN-CONTROLLED AI

AI recommends | Transparent rules prioritize | Humans authorize | Evidence proves closure

WHO USES IT

Citizens	Authorities	Contractors
Report + track	Review + assign	Accept + prove

EVIDENCE ENTERING THE SYSTEM

Photo • confirmed location • description

Optional private contact • planned WhatsApp channel

IMMEDIATE CITIZEN OUTCOME

Public case ID + status timeline + SLA visibility

No citizen account required; reporter identity stays out of public records

JUDGE LENS

Not another complaint form

A complete operating and accountability workflow

1 Report
Citizen submits evidence and confirms the location.

2 AI triage
Gemini recommends category, severity, department, confidence, and duplicate hints.

3 Review gate
Low confidence or ambiguity is routed to an officer for correction.

4 Prioritize
Deterministic rules score severity, safety, citizen signals, age, and location impact.

5 Assign
Authority selects an eligible contractor and locks the SLA and budget cap.

6 Prove work
Contractor uploads completion evidence and a work report.

7 Verify + track
Authority approval creates verified closure and a public accountability receipt.

EXPERIENCE LAYER

React 19 + Vite

- Citizen reporting flow
- AI review preview
- Authority operations dashboard
- Contractor work hub
- Public map + case tracking
- Before/after proof viewer

RESPONSIVE WEB

APPLICATION + DECISION LAYER

FastAPI REST backend

AI service
Gemini Vision + deterministic fallback

Workflow engine
state transitions + human review

Priority engine
bounded, explainable score / 100

Accountability
receipt + audit-chain summary

Lifecycle
SLA, assignment, proof, closure

DATA + EXTERNAL SERVICES

- **MongoDB** — complaints, users, offers, audit events
- **Cloudinary** — sanitized evidence storage
- **Gemini** — multimodal issue analysis
- **Stripe-compatible** — controlled payment release workflow
- **OpenStreetMap** — maps and location context

Runtime modes are shown as live, fallback, or demo.

TRUST + CONTROL LAYER

- Human review for uncertain AI
- Role-based access + CSRF protection
- PII screening + private reporter contact
- Image validation + metadata removal
- Idempotent complaint and payment actions
- Append-only audit events
- Contractor proof + authority approval

NO AI-AUTHORIZED CLOSURE

WHAT MAKES URBANFIX JUDGE-READY

WORKING
Deployed full-stack MVP with citizen, authority, contractor, and public views.

MEANINGFUL AI
AI structures messy evidence; confidence and reasoning remain visible.

EXPLAINABLE
Rules - not a black box - determine urgency and expose the score.

ACCOUNTABLE
A case is not closed until work evidence is inspected and approved.

HONEST
Demo records and fallback integrations are clearly labeled.

LIVE DEMO urbanfixpk.vercel.app

GITHUB github.com/asadali552/CivicPulse

Current MVP: web reporting | Planned: WhatsApp reporting + municipal integrations